

MESTTS SESSION REPORT

Multimodal Eye–Speech–Typing Temporal Synchronization System

Participant ID	p20	Session ID	1adff5d2-176d-47
Date / Time	2026-04-12 10:07	Sentences Scored	3

DYSLEXIA RISK ASSESSMENT

HIGH

Risk Score: 0.5500

EVS Gaze Span	Calibration Quality	Data Completeness	Sentences Scored
0.1482	56.2%	100.0%	3

Extracted Feature Vector

F01_evs_speech_std_ms 2196.3825	F02_mean_evs_speech_ms 3391.1333	F03_evs_speech_skew 0.1180	F04_speech_inversion_ 0.0000
F05_evs_typing_std_ms 2114.6600	F06_mean_fixation_dur. 169.1830	F07_std_fixation_dura. 11.8111	F08_regression_rate 0.2778
F09_regression_count_ 6	F10_fixation_outlier_ 1.1489	F11_typing_inversion_ 0.0000	F12_mean_correction_r. 0.0000
F13_max_corrections 0	F14_stg_std_ms 816.1022	F15_stg_negative_rate 0.1667	F16_corr_evs_speech_f. -0.0861
F17_corr_evs_typing_l. -0.1654			

IMPORTANT: This system produces a behavioral risk indicator only. It does not constitute a clinical diagnosis of dyslexia or any other condition. Results must be interpreted by a qualified specialist in the context of a comprehensive assessment protocol.

Interpretation

Overall Assessment

Significant markers of reading difficulty detected — clinical follow-up strongly advised.

The participant's score of 0.550 is in the high-risk band. Multiple behavioural indicators were present across sentences. A qualified specialist should evaluate these results in the context of a full assessment.

Data Quality

EVS Gaze Span — ACCEPTABLE

Span of 0.1482 is workable but on the lower side. Minor inaccuracies in word mapping are possible. Results should be interpreted with slight caution.

Calibration Quality — MODERATE

56% quality suggests the calibration was acceptable but not ideal. Word-level mapping may have minor errors.

Data Completeness — HIGH

100% of word slots had both gaze and speech data captured. The session data is highly reliable.

Key Reading Behaviour Indicators

Eye-Voice Span (F02)

The eye typically looked ahead of spoken words by ~3391 ms. This is normal — fluent readers pre-scan upcoming words.

EVS Consistency (F01)

EVS variability was high (2196 ms std). Erratic pacing — gaze and speech were poorly synchronised. This is a notable dyslexia marker.

Mean Fixation Duration (F06)

Mean fixation of 169 ms is in the typical range (150-280 ms). The participant paused a normal amount of time on each word.

Regression Rate (F08)

Regression rate of 27.8% is elevated. Frequent backward eye movements suggest difficulty decoding — a key indicator in dyslexia screening.

Gaze-Speech Correlation (F16)

Gaze-speech correlation of -0.086 is very low or negative. The participant's eye movements and speech were poorly synchronised.

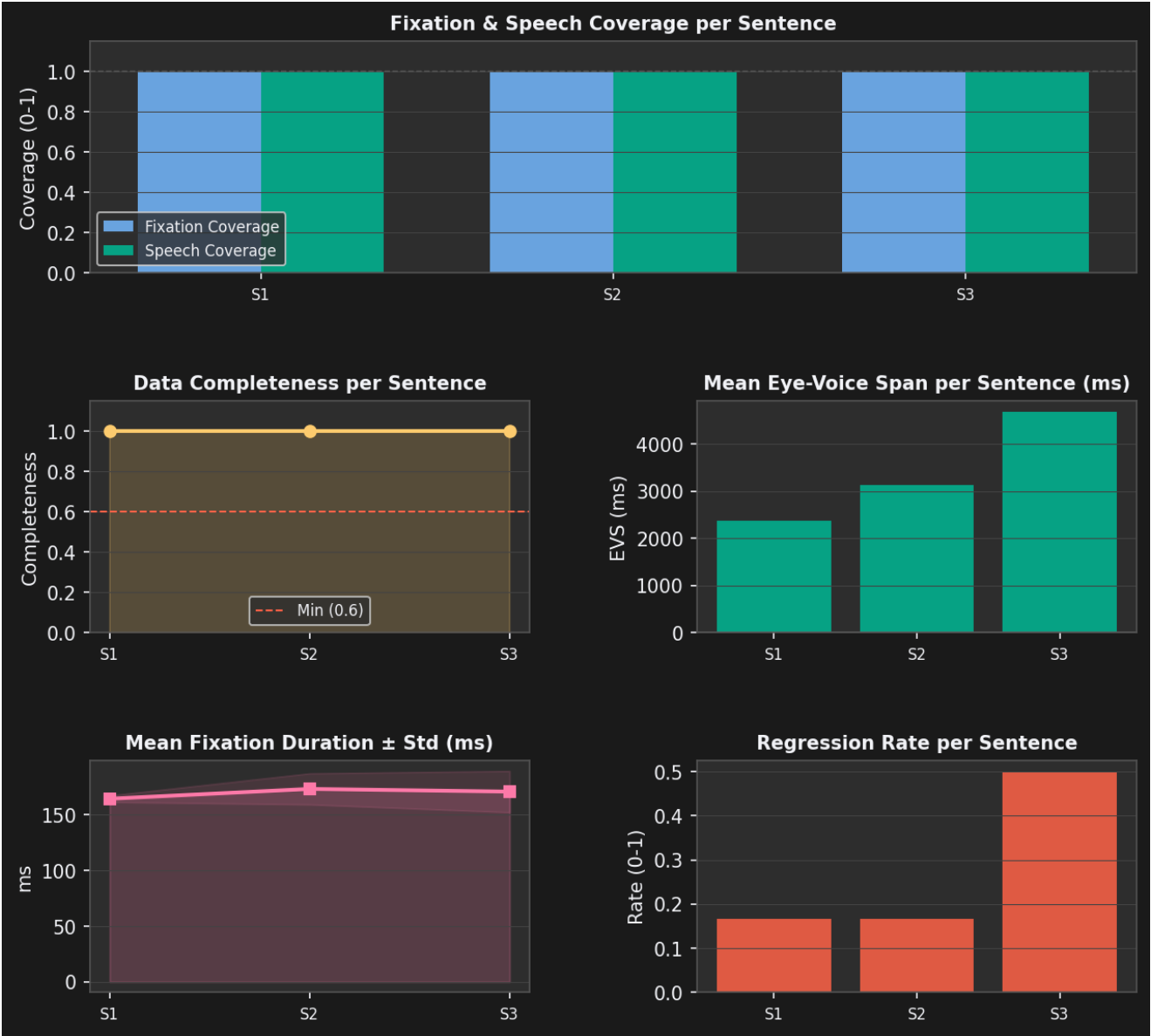
What to Do Next

Next Steps

These results warrant prompt referral to a qualified specialist — educational psychologist, speech-language therapist, or dyslexia specialist. Do not rely on this tool alone for diagnosis. Early structured literacy intervention significantly improves outcomes.

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Per-Sentence Analysis



Typing Analysis



Words Typed	Typing Coverage	Total Corrections	Mean STG	Mean EVS Typing
20 / 20	100%	0	1411 ms	4948 ms

Feature Vector

Feature	Value	Description
F01_evs_speech_std_ms	2196.3825	EVS Speech variability (std, ms)
F02_mean_evs_speech_ms	3391.1333	Mean EVS Speech (ms)
F03_evs_speech_skew	0.1180	EVS Speech skewness
F04_speech_inversion_rate	0.0000	Speech inversion rate (voice leads eye)
F05_evs_typing_std_ms	2114.6600	EVS Typing variability (std, ms)
F06_mean_fixation_duration_ms	169.1830	Mean fixation duration (ms)
F07_std_fixation_duration_ms	11.8111	Fixation duration std (ms)
F08_regression_rate	0.2778	Regression rate (re-reads / word)
F09_regression_count_total	6	Total regression count
F10_fixation_outlier_ratio	1.1489	Fixation outlier ratio (max/mean)
F11_typing_inversion_rate	0.0000	Typing inversion rate
F12_mean_correction_rate	0.0000	Mean correction rate (backspaces/word)
F13_max_corrections	0	Max corrections in any single word
F14_stg_std_ms	816.1022	Speech-Typing Gap std (ms)
F15_stg_negative_rate	0.1667	STG negative rate
F16_corr_evs_speech_fixation	-0.0861	Gaze-speech correlation (Pearson r)
F17_corr_evs_typing_latency	-0.1654	EVS-typing latency correlation

